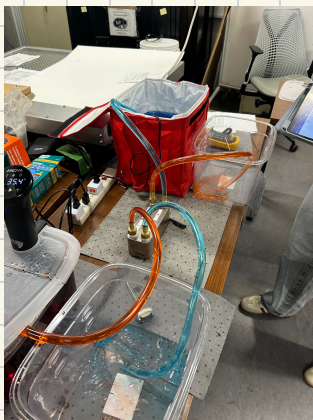
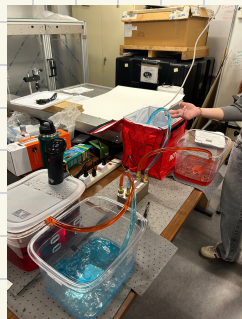


Heat Exchanger Lab

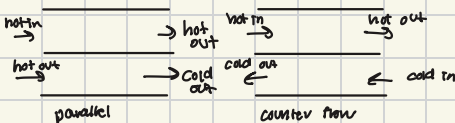
Photos of set up



Explanation of set up

The device is a small water to water heat exchanger. One stream carries hot water from the hot reservoir, and the other stream carries cold water from the cold reservoir. The two streams never mix, but heat is transferred by conduction through the metal separating the channels and convection between the fluids and metal surface. We did different types of flow by altering our set up: parallel flow, counter flow, and altered the mass flow rate.

system diagram



Data during experiment

① $T_{H,i} : 43.3^\circ\text{C}$ $T_{C,i} : 9.3^\circ\text{C}$
 33.0°C 9.3°C

parallel

② $T_{H,i} : 35.4^\circ\text{C}$ $T_{C,i} : 19.6^\circ\text{C}$
 25.6°C 16.7°C

counterflow

③ $T_{H,i} : 35.9^\circ\text{C}$ $T_{C,i} : 18.1^\circ\text{C}$
 26.5°C 21.4°C

parallel w/ changed mass flow rate

mass balance:

$$\dot{m}_{H,i} = \dot{m}_{H,out} = \dot{m}_H$$

$$\dot{m}_{C,i} = \dot{m}_{C,out} = \dot{m}_C$$

Entropy Balance:

$$\sum \dot{m}_i s_{i,in} - \frac{\dot{Q}_{av}}{T_0} = \sum \dot{m}_i s_{i,out} + \dot{S}_{gen} \quad \dot{S}_{gen} \geq 0$$

$$\dot{m}_H c_p \ln \left(\frac{T_{H,out}}{T_{H,in}} \right) + \dot{m}_C c_p \ln \left(\frac{T_{C,out}}{T_{C,in}} \right) - \frac{\dot{Q}_{av}}{T_0} = \dot{S}_{gen} \geq 0$$

$$T_{C,out} > T_{H,out}$$

Energy Balance

$$\sum \dot{m} \left(h + \frac{V^2}{2} + gz \right)_{in} + \dot{Q} - \dot{W} = \sum \dot{m} \left(h + \frac{V^2}{2} + gz \right)_{out}$$

$$\Delta KE = 0 \quad \Delta PE = 0$$

$$\dot{m}_H h_{H,i} + \dot{m}_C h_{C,i} + \dot{Q}_{av} = \dot{m}_H h_{H,out} + \dot{m}_C h_{C,out}$$

$$\dot{m}_H c_p (T_{H,i} - T_{H,out}) = \dot{m}_C c_p (T_{C,out} - T_{C,i}) + \dot{Q}_{av}$$

$$\dot{m}_H (T_{H,i} - T_{H,out}) = \dot{m}_C (T_{C,out} - T_{C,i})$$

Effect of Design/operating changes

we slowed down the mass flow rate on the hot temperature side

$$R_H: 35.3^\circ \rightarrow 21.4^\circ \text{C}$$

$$R_C: 13.1^\circ \rightarrow 26.5^\circ \text{C}$$

From energy balance: $\dot{m}_h(T_{h,m} - T_{h,out}) = \dot{m}_c(T_{c,out} - T_{c,m})$. So since we slowed down the mass flow rate on the hot side in order to keep the energy balance there must be a larger temperature drop on the hot side.

Specific Analysis

Alfa Laval T2 Plate Heat Exchanger

This is a compact plate heat exchanger that is used in HVAC and industrial cooling. The T2 uses a stack of metal plates to increase heat transfer.

Specifications:

Model: Alfa Laval T2

Heat transfer area: $0.04 - 0.06 \text{ m}^2$

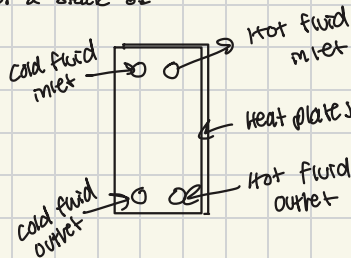
Typical plate count: 10-50

Max operating temperature: 180°C

Max operating pressure: 1.6 MPa

Common fluids: water, glycol mixtures, oils

How the Alfa Laval T2 works:

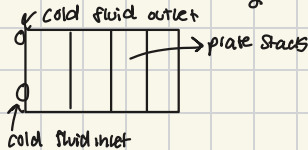


Flow path

1. Hot & cold fluids enter alternate plate channels
2. Each plate is gasketed, forcing the fluids into zig-zag pathways
3. High turbulence enhances convection coefficients by 2-5x compared to smooth channels

Heat transfer mechanism:

1. Hot fluid converts heat to the stainless steel plate surface
2. Heat conducts through the corrugated metal plate
3. Cold fluid converts heat away from the opposite side



Analysis using real device numbers

Assume:

Hot side: $35.3^\circ \text{C} \rightarrow 21.4^\circ \text{C}$

Cold side: $13.1^\circ \text{C} \rightarrow 26.5^\circ \text{C}$

Fluids: water

Hot side flow rate: 0.8 kg/s

Cold side flow rate: 0.9 kg/s

$c_p = 4180 \text{ J/kg}\cdot^\circ\text{C}$

Energy Balance:

Hot side heat loss:

$$\dot{Q}_h = \dot{m}_h c_p (T_{h,m} - T_{h,out}) = (0.8)(4180)(35.3 - 21.4) = 74,236.8 \text{ J/s} = 74.23 \text{ kW}$$

Cold side:

$$\dot{Q}_c = \dot{m}_c c_p (T_{c,out} - T_{c,m}) = (0.9)(4180)(26.5 - 13.1) = 80,410.8 \text{ J/s} = 80.4 \text{ kW}$$

Entropy Generation

$$\dot{S}_{gen} = \dot{m}_h c_p \ln\left(\frac{T_{h,out}}{T_{h,m}}\right) + \dot{m}_c c_p \ln\left(\frac{T_{c,out}}{T_{c,m}}\right)$$
$$\dot{S}_{gen} = (0.8)(4180) \ln\left(\frac{21.4}{35.3}\right) + (0.9)(4180) \ln\left(\frac{26.5}{13.1}\right)$$

$$\dot{S}_{gen} = 970.80 \text{ W/K}$$

$$\rightarrow 0.970 \text{ kW}$$

Comparison to Experimental Lab

- The small lab showed uneven heating of the block so it wasn't perfectly isovolumetric
- The heat exchanger lab shows "parallel & counterflow" which matches the known differences for plate heat exchangers like ΔT_2